Paper Time

[90 minutes]

CS408 – Human Computer Interaction

CS408 (Final Exam Sample Paper)

MCQs

Take an assessment.

Question#1
_____ in the cost of computing resources have resulted from new technological advances i.e. the development of the silicon chip. A. Dramatic decrease B. Dramatic increase C. Neither increase nor decrease D. Less decrease
Question#2
Human-computer interface specialists are _____ and software engineers are_____. A. User-centered, System-centered B. System-centered, User-centered C. System-centered, Design-centered D. User-centered, Design-centered
Question#3
_____ occurs to destroy the physical health, the emotional stability, and if left unchallenged, the economic strength of whatever population it strikes. A. Computer Rage B. Computer virus C. Denial of service (DoS) attack D. Worm

Question#4

_____ designed the first object-oriented programming language in the 1970s called Smalltalk.

A. Alan Kay

B. Alan Turing

C. Elon musk

D. Dennis Ritchie

Question#5

_____ refers to defining and designing tools and artifacts for different work and environments to suit the capabilities and capacities of users.

A. Ergonomics

B. Mnemonics

C. Economics

D. Semantics

Question#6

Experiential and Reflective are the two _____ distinguished by Norman in 1993.

A. Modes of Cognition

B. Modes of Recognition

C. Modes of Idiotic

D. Modes of irrationality

Question#7

Color is usually regarded as being made up of_____.

A. Hue, intensity, and saturation

B. Rods, Cones, and Fovea

C. Red, White, and black

D. Blue, Yellow, and white

Question#8

Green, orange, and purple colors are examples of _____.

A. Secondary Colors

B. Primary Colors

C. Tertiary Colors

D. Hybrid Colors

Question#9

_____ are any three colors, which are side by side on a 12-part color wheel and where one of the three colors predominates.

A. Analogous colors

B. Complementary colors

C. Primary Colors

D. Hybrid Colors

Question#10

Distracted from working when we can hear music or voices in the next room is an example of _____.

A. Involuntary attention

B. Voluntary attention

C. Focusing attention

D. Divided Attention

Question#11

_____ derives the logically necessary conclusion from the given premises.

A. Deductive reasoning

B. Inductive reasoning

C. Abdicative reasoning

D. Local Reasoning

Question#12

Norman has categorized errors into two main types which are given as _____.

A. Slips and mistakes

B. Syntax and semantic

C. Logical and semantic

D. Logical and syntax

Question#13

_____ allow us to control the computer by simply looking at it, some systems require us to wear special glasses or a small head-mounted box to use it.

A. Eye-gaze

B. Stylus

C. Light pen

D. Track point

Question#14

_____ is the default interface style for the majority of interactive computer systems in use today, especially in the PC and desktop workstation arena.

A. The WIMP Interfaces

B. Three-dimensional interfaces

C. Form fills and spreadsheets

D. Command line interface

Question#15

_____ has the intention to create a computing infrastructure that permeates our physical environment so much that we do not notice the computer any longer.

A. Ubiquitous computing

B. Sensor-based and context-aware interaction

C. Computer-supported cooperative work

D. Multi-modality

Question#16

_____aspect gives hints and tips about using and creating user interface idioms, like dialog boxes and push buttons.

A. Tactical

B. Strategic

C. Dominant

D. Logical

Question#17

_____ was derived from empirical work on understanding how designers tackled HCI design problems, it is a very flexible process with evaluation at its core.

A. Star Lifecycle model

B. Usability engineering lifecycle

C. Spiral Model

D. Rapid Application Development (RAD) model

Question#18

_____are the perfect perpetual intermediate idiom as they say nothing about scope, purpose and meaning; they only state function in the briefest of idioms.

A. Tooltips

B. Tactical

C. Strategic

D. Form fills and spreadsheets

Question#19

_____ can tell you about what, how, and why in rich, multivariate detail which helps us understand the domain, context, and constraints of a product in different, more useful ways.

A. Qualitative research

B. Quantitative research

C. Empirical Research

D. Statistical Research

Question#20

Stakeholder interviews, Subject matter experts (SME) interviews, User and customer interviews, and Literature review are the different types of _____.

A. Qualitative research

B. Quantitative research

C. Empirical Research

D. Statistical Research

Question#21

_____ requirements can include development timelines, regulations, pricing structures, and business models.

A. Technical

B. Business

C. Customer

D. Partner

Question#22

Containers for objects that must be compared or used together should be _____ to each other.

A. Adjacent

B. Detached

C. Separate

D. Disconnected

Question#23

_____ principles help define the look and feel of interfaces.

A. Conceptual-level

B. Interaction-level

C. Interface-level

D. Organization-level

Question#24

The posture of your _____ tells you much about its behavioral stance, which, in turn, dictates many of the important guidelines for the rest of the design.

A. Interface

B. Tool

C. Desktop

D. Window

Question#25

Many developers imagine the ideal interface to be a _____ conversation with the user.

A. One-way

B. Two-way

C. Three-way

D. Four-way

Question#26

The only way to determine whether a function or behavior is excise is by comparing it to

the _____.

A. Tester goals

B. Analyst goals

C. Developer goals

D. User's goals

Question#27

Measuring user performance to compare two or more designs has been the bedrock of

_____ testing.

A. Usability

B. Unit

C. Integration

D. Acceptance

Question#28

An ethnographic study in _____ homes would be more appropriate.

A. Testers’

B. Designers’

C. Engineers’

D. Users’

Question#30

_____ names are the street signs of the Web.

A. Page

B. Client

C. Book

D. Nick

Question#31

To implement rollovers in tabs or button bars, each button needs to be a separate _____.

A. Graphic

B. Pixel

C. Color

D. Size

Question#32

_____ ads take up too much space.

A. TV

B. Banner

C. Party

D. Commercial

Question#33

On the _____ side, we are mainly concerned with tasks-the steps involved in a process and how people think about completing them.

A. Hardware

B. Customer

C. Software

D. Developer

Question#34

A more experienced computer user might visualize undo as a storage facility for _____ data.

A. Deleted

B. Updated

C. Old

D. New

Question#35

If the user decides to _____ the document, he can just click on it and edit it in place.

A. Delete

B. Update

C. Rename

D. Hide

Question#36

The human _____ is a superb pattern-processing computer.

A. Kidney

B. Liver

C. Brain

D. Heart

Question#37

The most important thing to remember about _____ is that simple is better.

A. Tools

B. Grids

C. Fonts

D. Headers

Question#38

A _____ computer can be a useful alternative but it is more obtrusive and cumbersome, and its batteries need recharging every few hours.

A. Pocket

B. Super

C. Laptop

D. Hybrid

Question#39

For _____ the interviewer has a basic script for guidance.

A. Testing

B. Development

C. Consistency

D. Redundancy

Question#40

Most error message boxes are informing them _____ of the inability of the program to work flexibly.

A. Tester

B. Developer

C. Analyst

D. User

Subjective Questions (3 Marks)

Question#1

What is the fundamental difference between the approaches taken by software engineers and human-computer interaction specialists?

Answer:

Software engineers are **system-centered**, focusing on creating efficient and functional systems with technical precision. HCI specialists are **user-centered**, emphasizing usability, accessibility, and user satisfaction to create intuitive interfaces that meet user needs.

Question#2

While perceiving color it is usually regarded as being made up of three components, you are required to name these components.

Answer:

The three components of color perception are:

1. **Hue**: Type of color (e.g., red, blue).
2. **Saturation**: Intensity or purity of the color.
3. **Brightness/Value**: Lightness or darkness of the color.

Question#3

What is your understanding regarding the term “Ubiquitous Computing”?

Answer:

Ubiquitous Computing refers to integrating computing devices seamlessly into daily life. These devices are interconnected, invisible, and work automatically in the background, making technology more accessible and convenient. Examples include IoT devices like smart homes and wearable tech.

Question#4

"Scenarios are stories about people and their activities." Justify this statement.

Answer:

Scenarios describe how users interact with a system to achieve goals, focusing on **real-life use cases**. They detail the user's tasks, challenges, and motivations, helping developers understand usability issues and design features aligned with user needs.

Question#5

Why Co-discovery is more expensive than single participant testing? Explain.

Answer:

Co-discovery involves two participants, making it costlier due to **double recruitment costs**. Sessions take longer as participants collaborate, and analyzing their interactions requires additional time and effort, increasing overall expenses

Question#6

What is use of text and color in visual interfaces? Explain.

Answer:

Text provides **clear information**, ensuring readability and organizing content with headings. Color enhances **visual appeal**, guides navigation, and communicates statuses (e.g., red for errors). Together, they improve usability and accessibility in interfaces.

Subjective Questions (5 Marks)

Question#1

What are the different types of techniques used for qualitative research?

Answer:

Qualitative research focuses on gathering non-numerical, in-depth data. Common techniques include:

1. **Interviews**: Direct, open-ended discussions with participants to explore their experiences and perspectives.
2. **Focus Groups**: Small group discussions to collect diverse opinions and ideas on a topic.
3. **Observation**: Watching participants in their natural environment to understand behaviors and interactions.
4. **Case Studies**: Detailed analysis of a specific individual, group, or situation.
5. **Document Analysis**: Reviewing written, visual, or audio materials to gain insights into the topic.

Question#2

Briefly describe the three types of memory or memory functions as part of the multi-store model of memory.

Answer:

The multi-store model of memory consists of three types of memory functions:

1. **Sensory Memory**: Captures brief sensory information (e.g., visual, auditory) for a few milliseconds. If not attended to, it fades quickly.
2. **Short-term Memory (STM)**: Holds information temporarily (about 20–30 seconds) and has limited capacity. Rehearsal helps transfer it to long-term memory.
3. **Long-term Memory (LTM)**: Stores information indefinitely with unlimited capacity. Information is retrieved from LTM when needed.

Question#3

To create a copy of a document, you can use several methods, depending on the type of document and the device you're using. Describe any three common ways to create a copy of a document.

Answer:

- **Copy-Paste Method**: Highlight the content, use the copy option (Ctrl + C), and paste it into a new document (Ctrl + V).
- **Save As**: Open the document, choose "Save As," and save it under a different name or location to create a duplicate.
- **Duplicate Option**: Many file managers allow creating duplicates directly by right-clicking on the file and selecting "Duplicate" or "Copy."

Question#4

Define ubiquitous computing. What is the goal of ubiquitous computing? Enlist any two examples of ubiquitous computing.

Answer:

Definition: Ubiquitous computing refers to embedding computing devices seamlessly into everyday objects, enabling them to interact with users and environments invisibly.
Goal: To make computing intuitive and accessible, enhancing convenience and user experience without requiring active user interaction.

Examples:

1. Smart home systems (e.g., automated lighting and temperature control).
2. Wearable devices (e.g., fitness trackers and smartwatches).